

SAE-3 : CARDIOFRÉQUENCEMÈTRE

ARTHUR-MALAN-A42-ESE-GEII

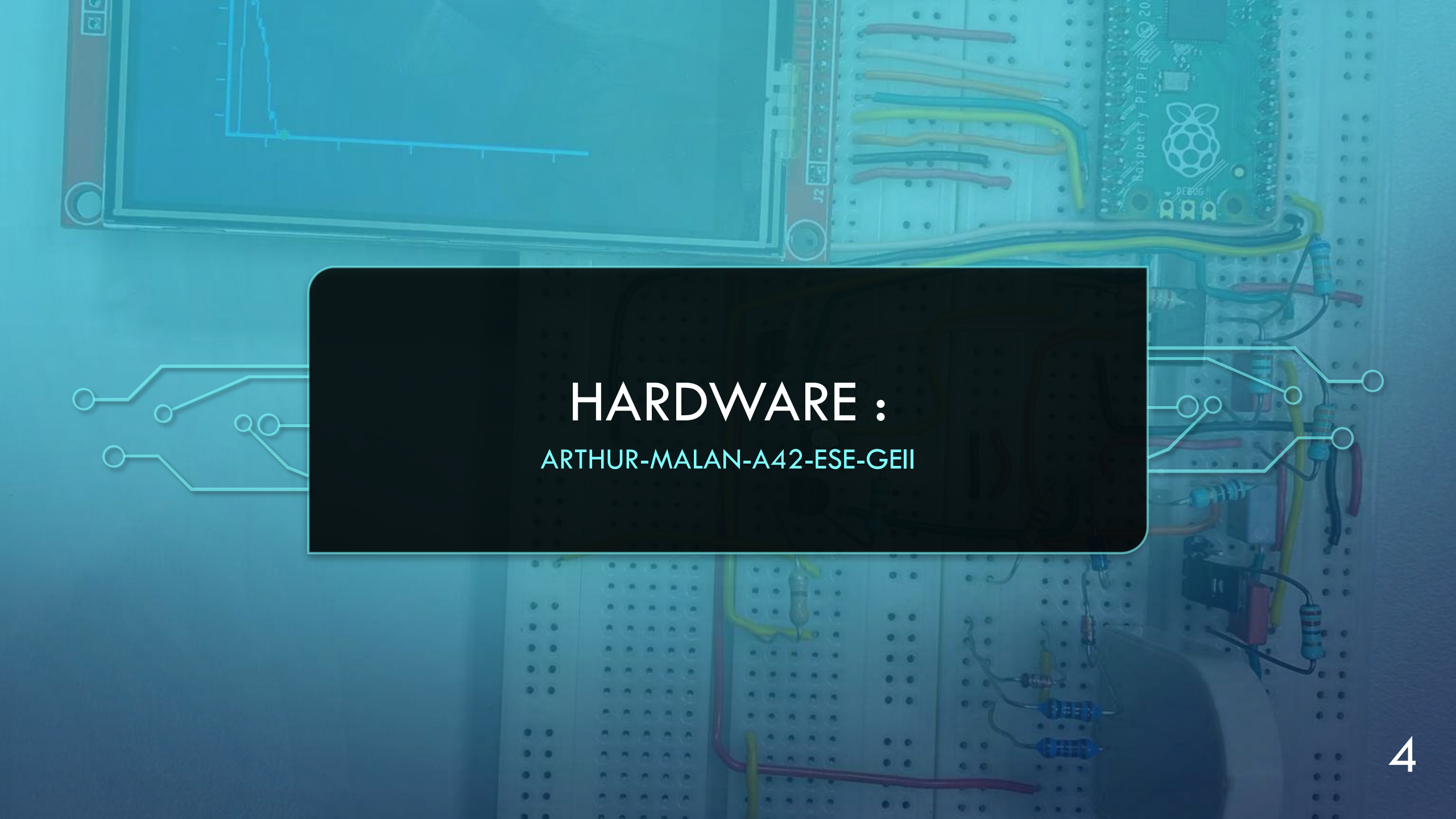


INTRODUCTION :

CIRCUITPYTHON, TP D'INFO2, GITHUB

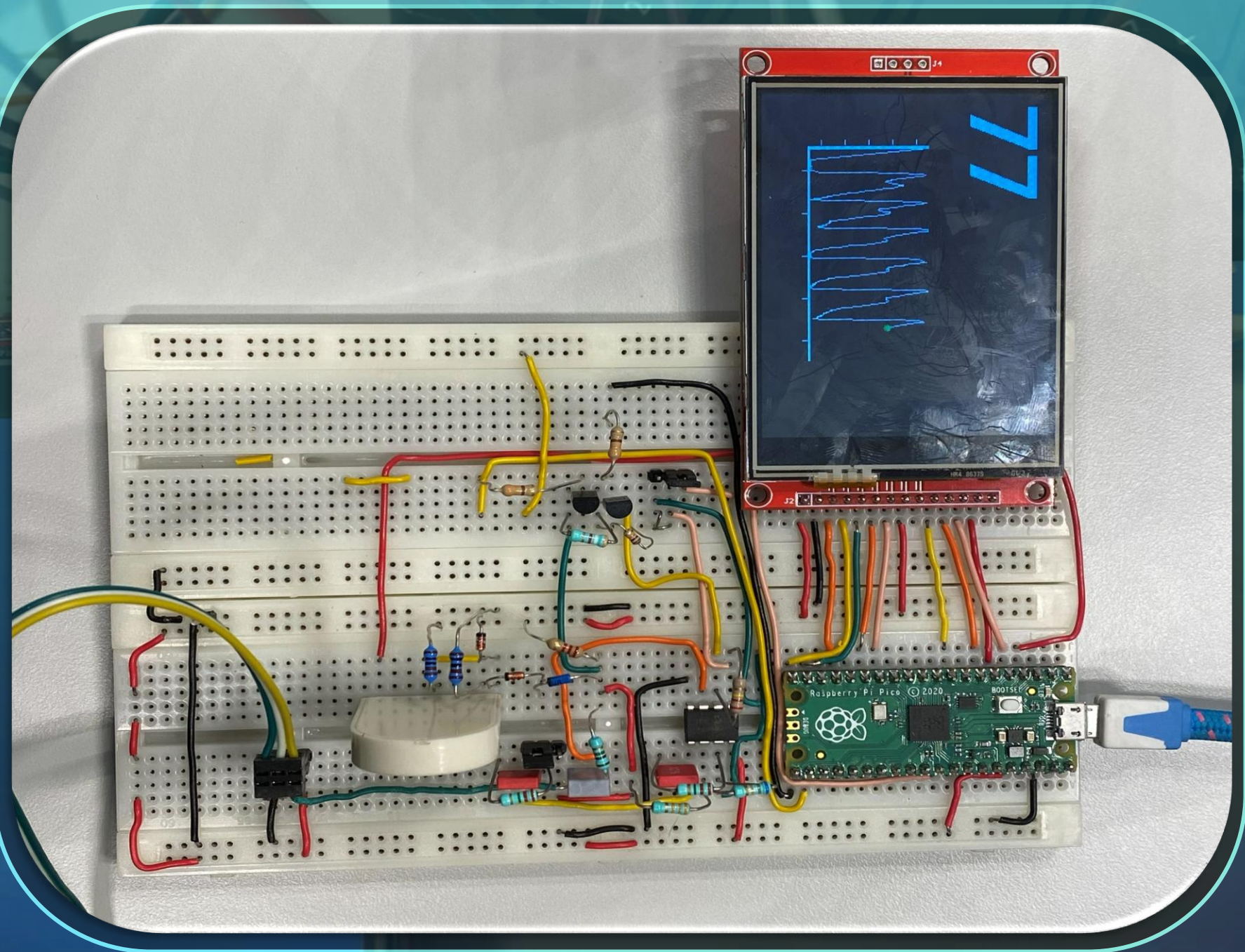
PLAN :

- Hardware
- Software
- Mode d'utilisation



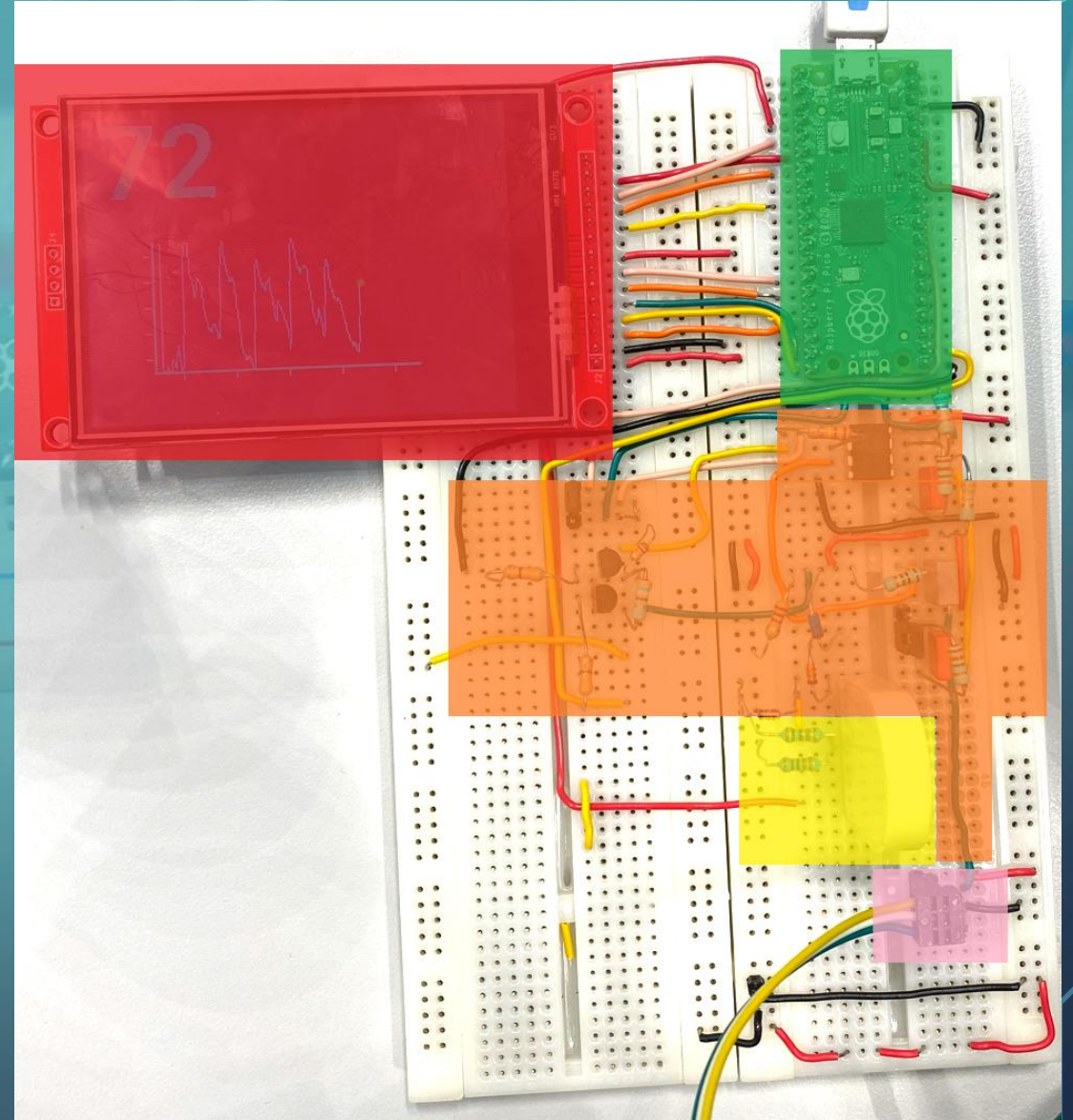
HARDWARE :

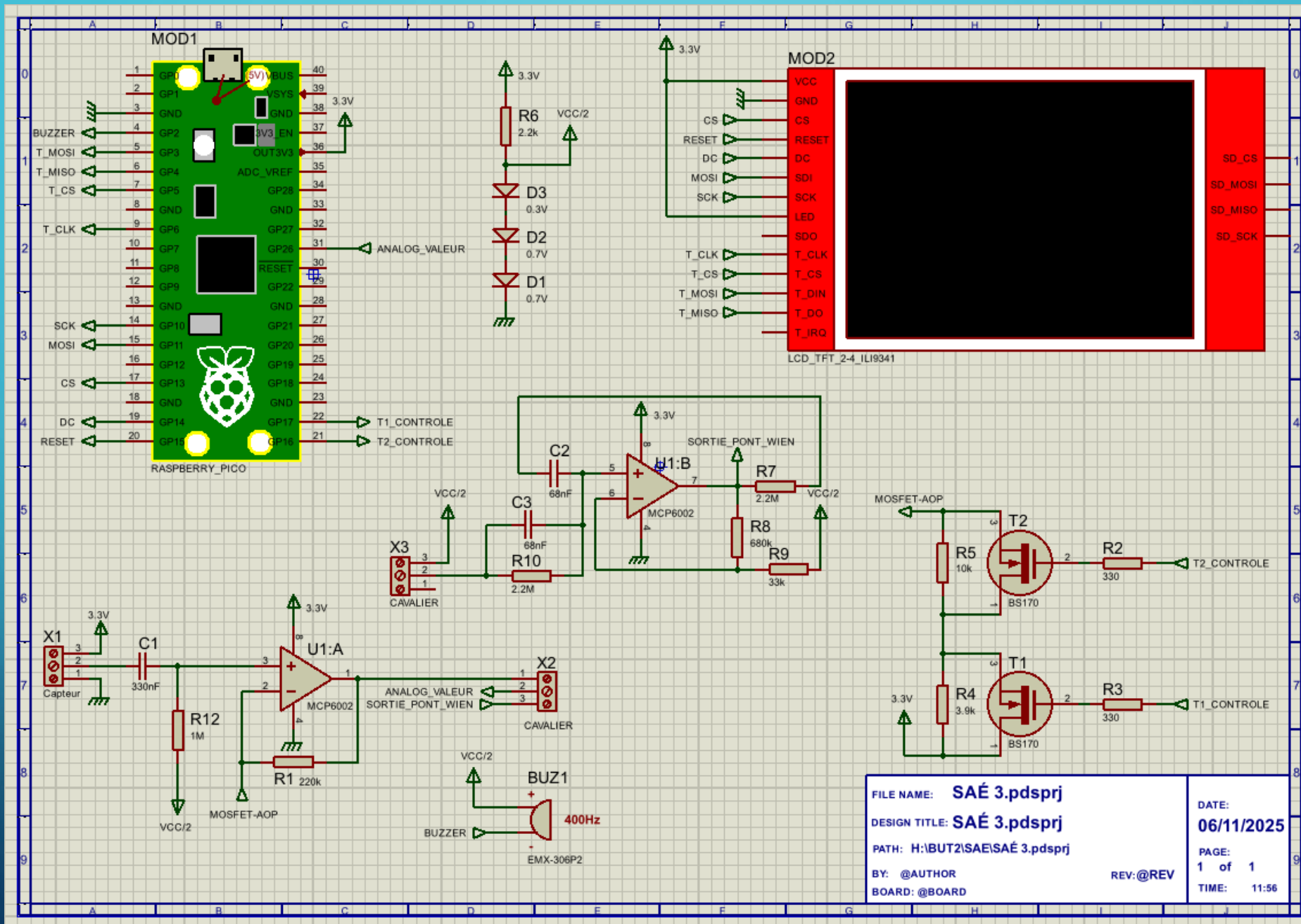
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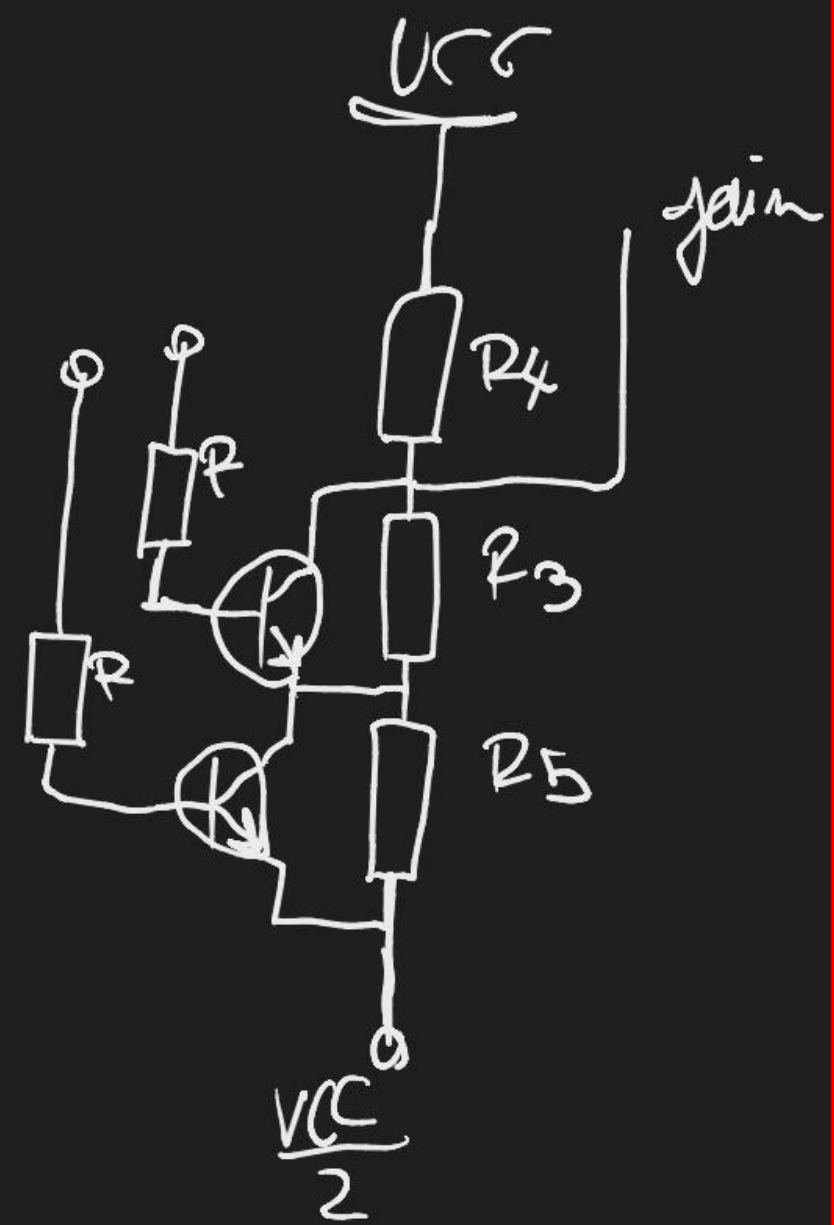
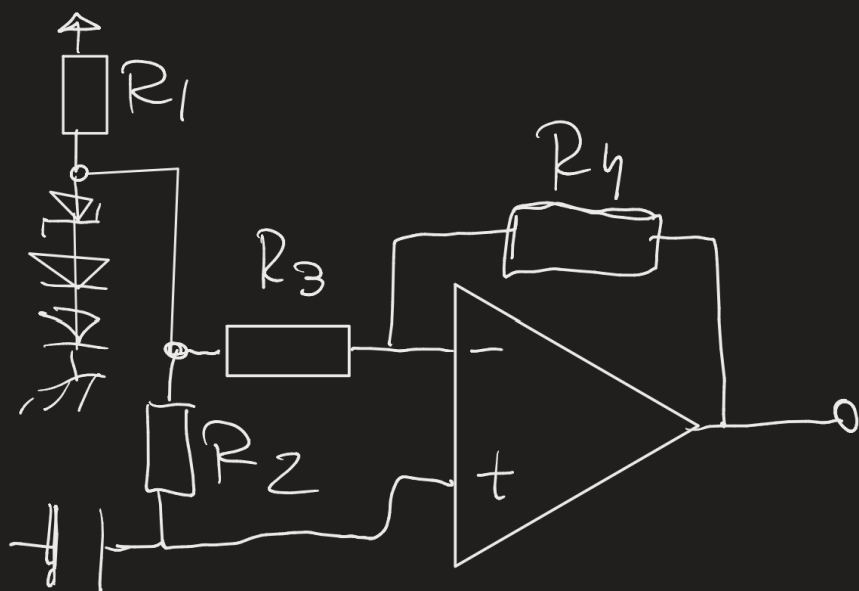
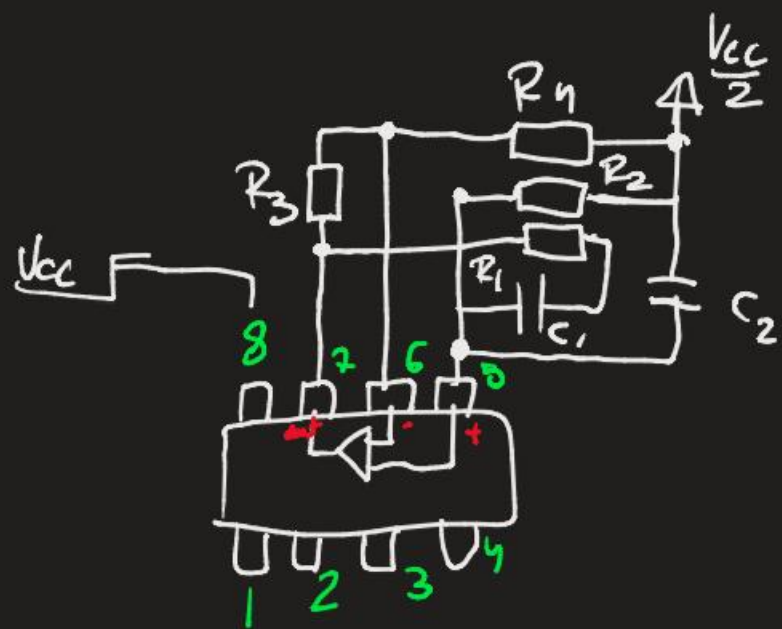


PARTIE DU SYSTÈME

- IHM
- Microcontrôleur
- traitement du signal
- Capteur
- Indicateur sonore (Buzzer)







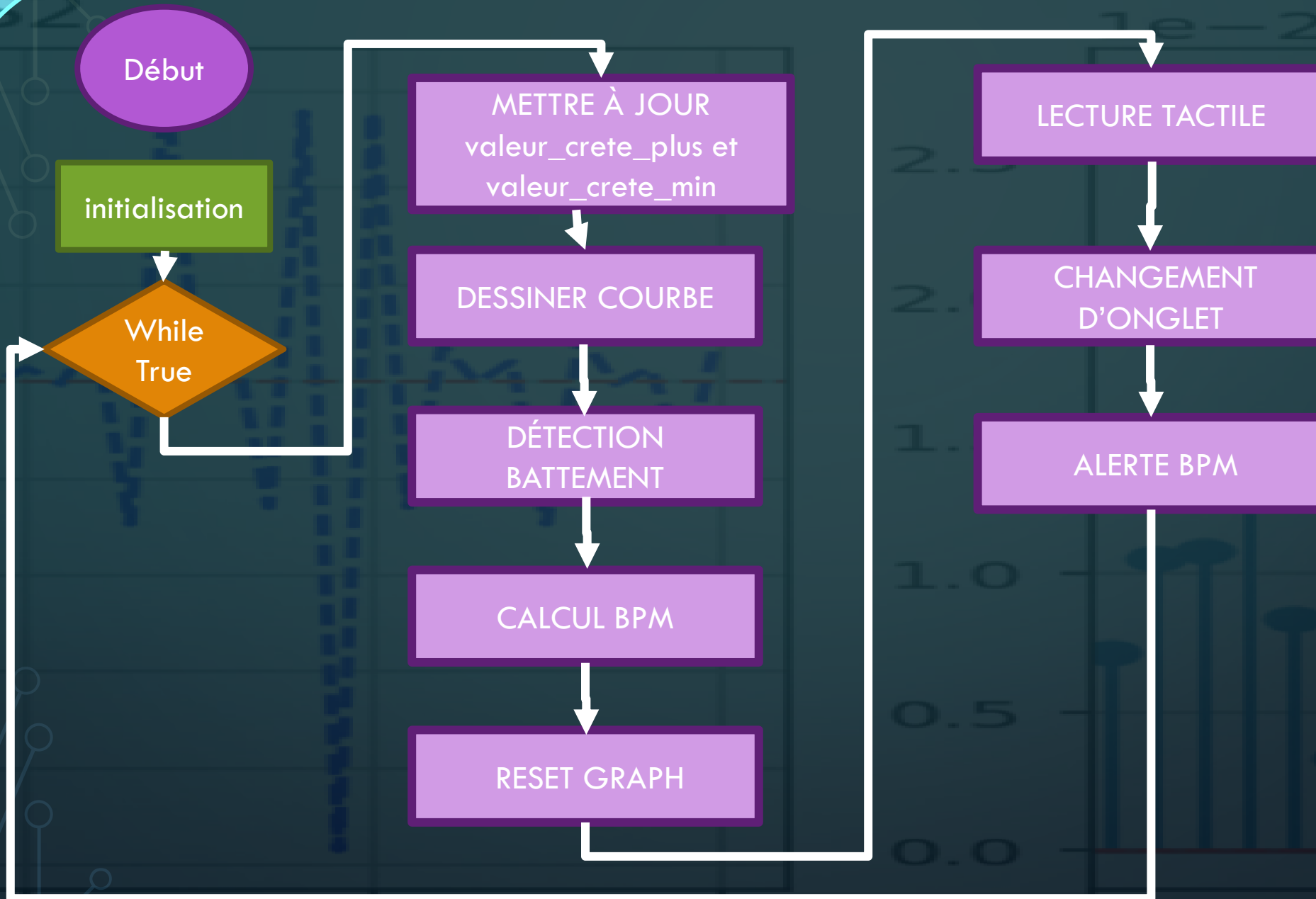

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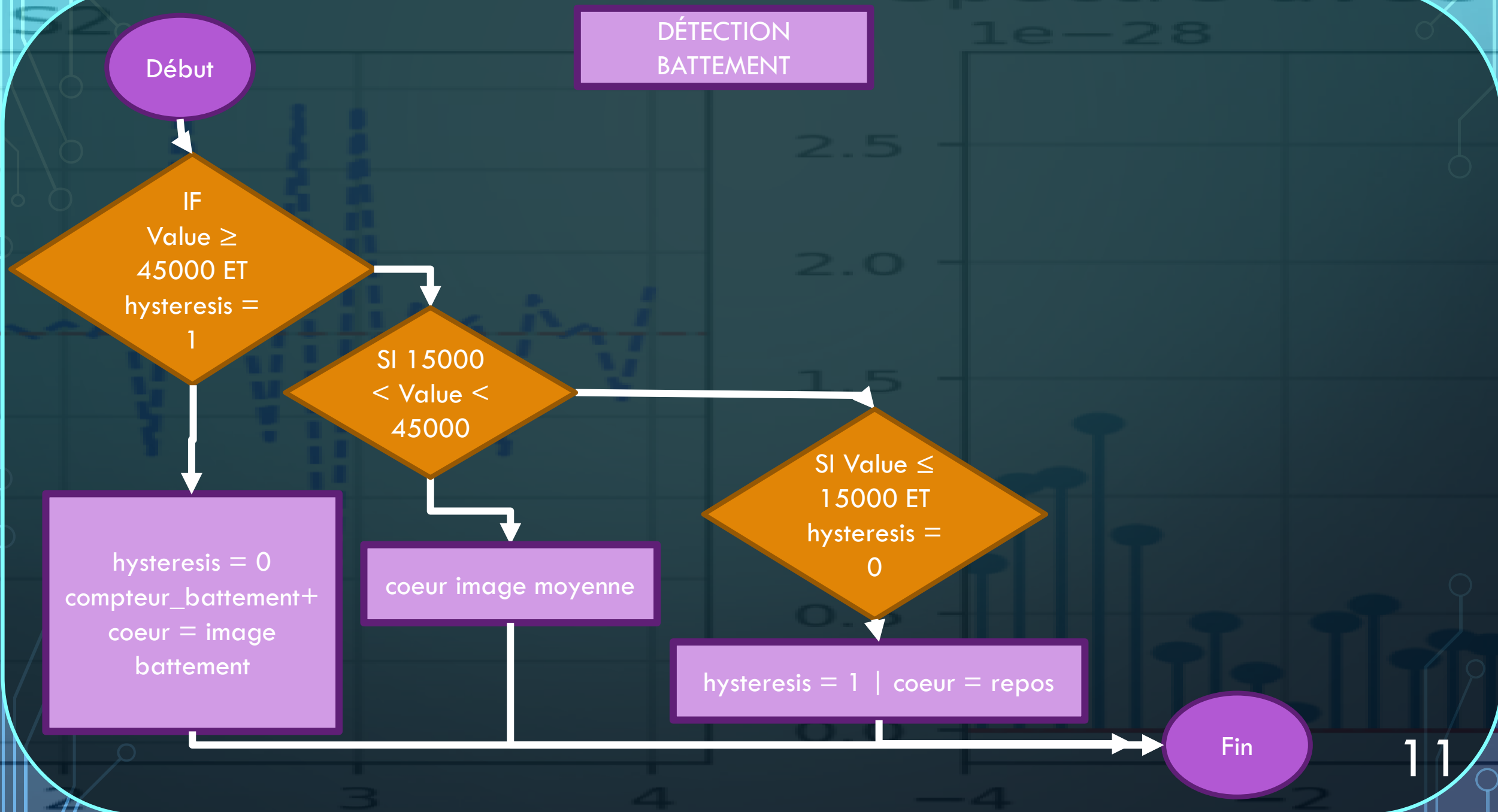
1 yrange=(0, 65536),
2 font_color = 0x000000,
3 tick_color=0x24FFF8,
4 )
5 groupe_elements.append(my_plane)
6
7 # Importe la police d'écriture
8 font_file = "fonts/Roboto-SemiBold-80.pcf"
9 BPM_text = label.Label(bitmap_font.load_font(font_file), text="Wait..", color=0x24FFF8,x=10, y=35, scale=1)
10
11 groupe_elements.append(BPM_text)
12
13 # Importe les images
14 fichier_image = open("images/rouage.bmp", 'rb')
15 image_rouage = displayio.OnDiskBitmap(fichier_image)
16 tuile_image_rouage =displayio.TileGrid(image_rouage, pixel_shader=displayio.ColorConverter(),x=240,y=20)
17 groupe_elements.append(tuile_image_rouage)
18
19 fichier_image_fleche_bas = open("images/fleche_bas.bmp", 'rb')
20 image_fleche_bas = displayio.OnDiskBitmap(fichier_image_fleche_bas)
21 tuile_image_fleche_bas =displayio.TileGrid(image_fleche_bas, pixel_shader=displayio.ColorConverter(),x=120,y=180)
22
23 fichier_image_fleche_haut = open("images/fleche_haut.bmp", 'rb')
24 image_fleche_haut = displayio.OnDiskBitmap(fichier_image_fleche_haut)
25 tuile_image_fleche_haut =displayio.TileGrid(image_fleche_haut, pixel_shader=displayio.ColorConverter(),x=120,y=10)
26 ecran.root_group = groupe_elements
27 # Initialisation du texte de la fréquence max
28 freq_max = 165
29 Freq_max_text = label.Label(bitmap_font.load_font(font_file), text=f"{freq_max}", color=0x24FFF8,x=80, y=120, scale=1)
30
31 adc = analogio.AnalogIn(board.GP26)
32
33 # Initialisation du PWM pour le buzzer
34 pwm = pwmio.PWMOut(board.GP2, frequency=500)
35 pwm.duty_cycle = 0
36
37 # varriable global
38 valeur_crete_plus = 0
39 valeur_crete_min = 65535

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SOFTWARE :

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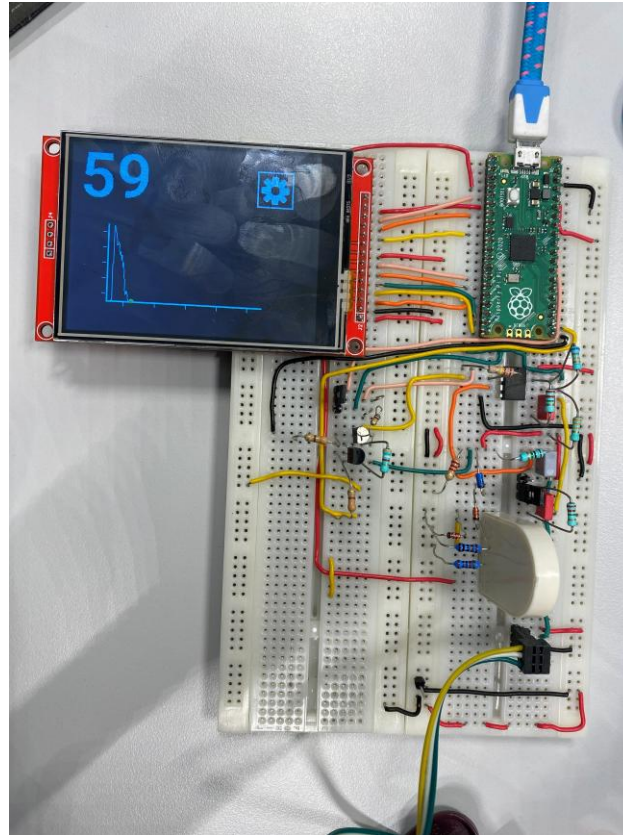
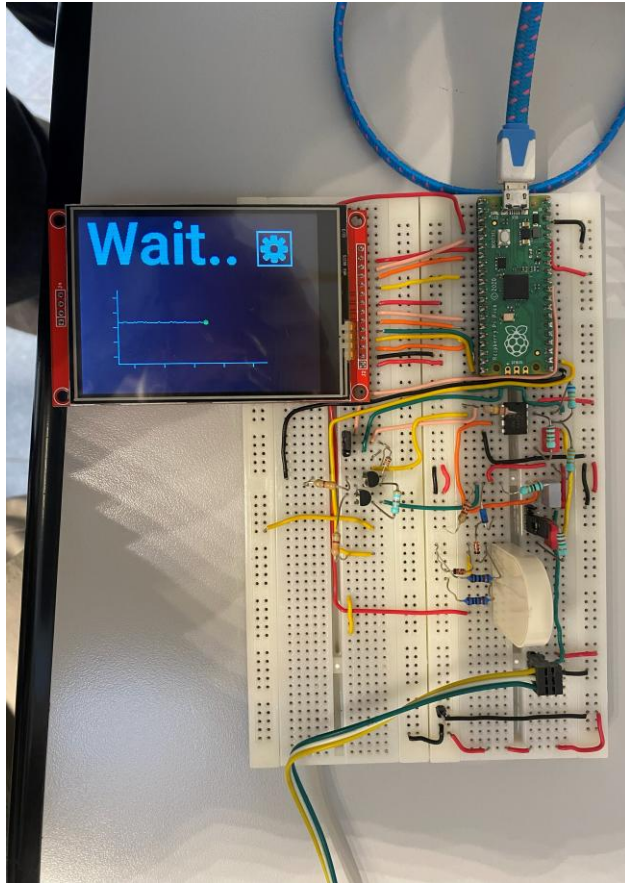
CALCUL BPM

$$BPM = \frac{4 \text{ battements}}{\text{durée (s)}} \times 60 = \frac{240}{\text{durée (s)}}$$

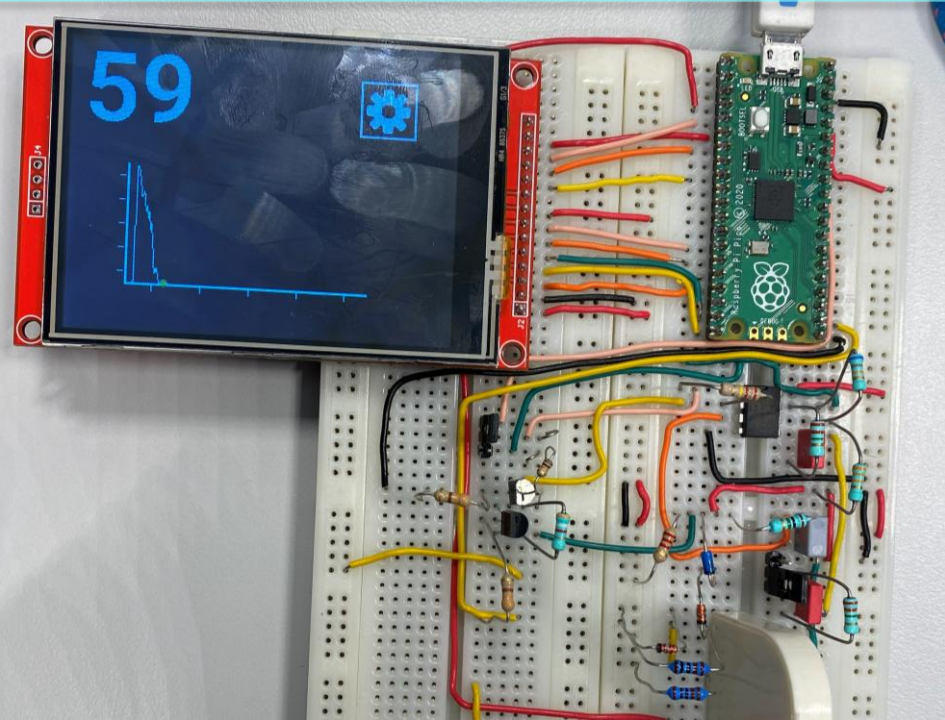
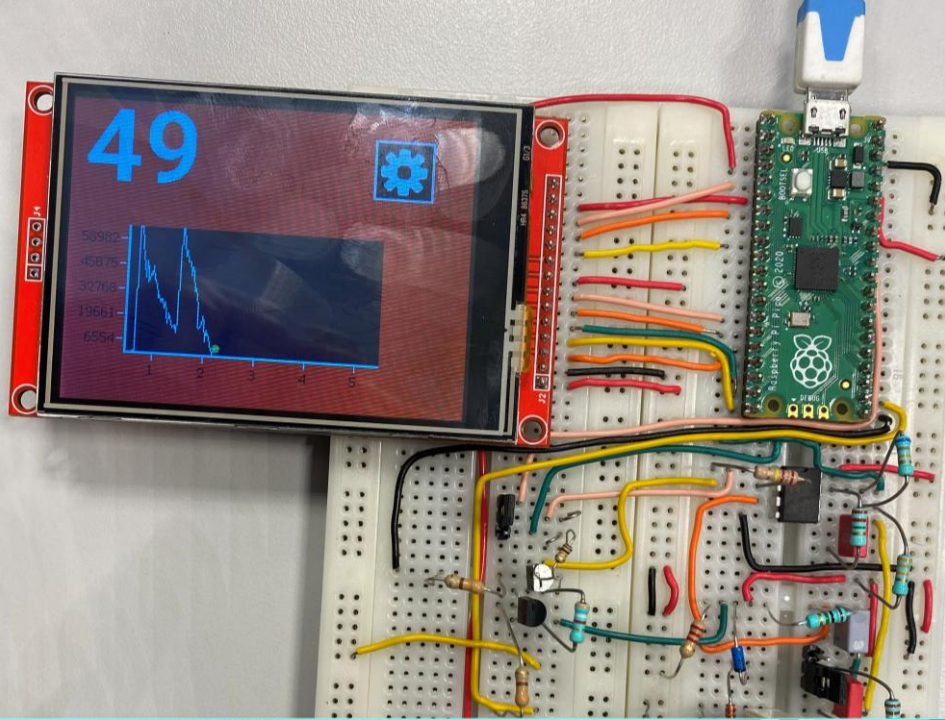


MODE D'UTILISATION :

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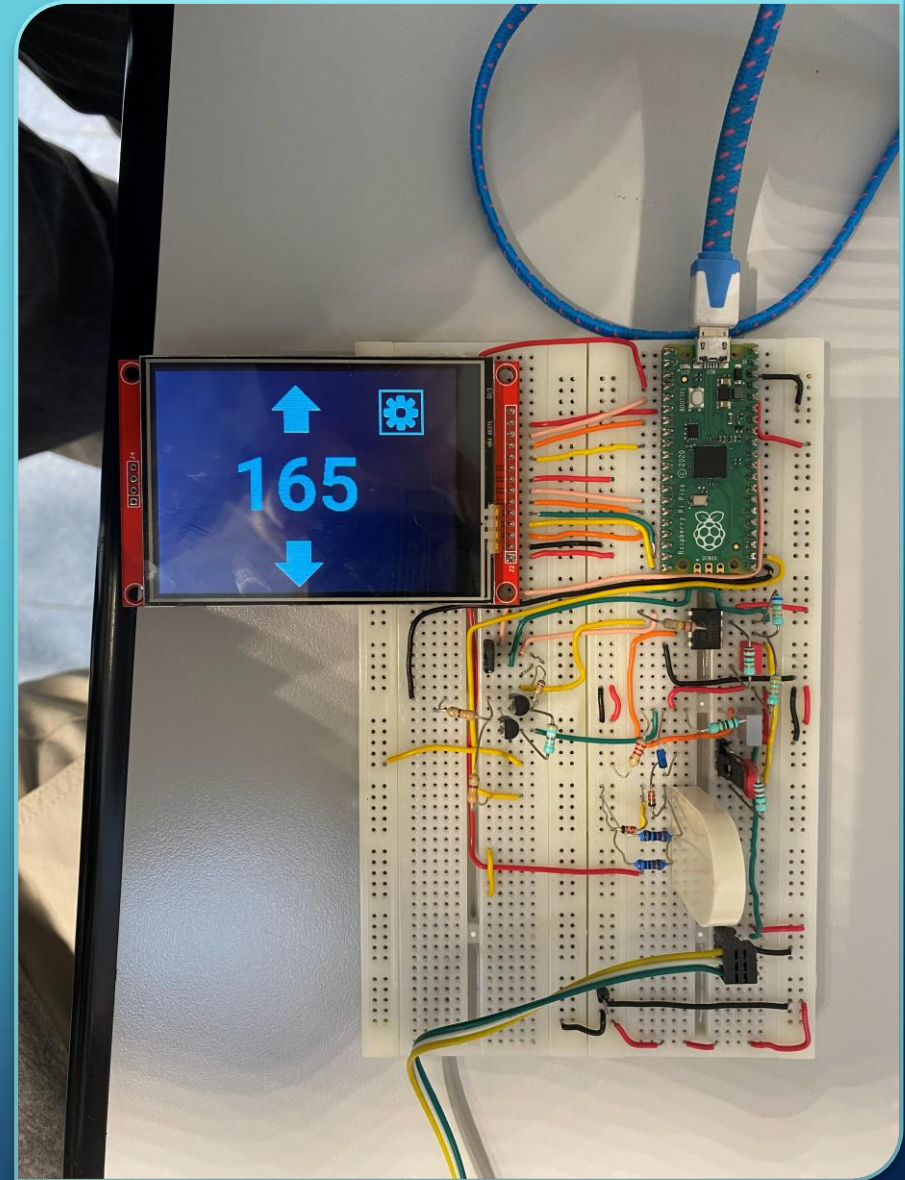
« WAIT... » AU DÉBUT
CAR LA MESURE A BESOIN
D'UN MINIMUM
D'ÉCHANTILLONNAGE
POUR DONNER
CORRECTEMENT LE
NOMBRE DE BATTEMENT.



ALERTE SI ONT PASSE AU
DESSUS DU SEUIL PRÉDÉFINI.

- Alerte aussi si on ne détecte plus de battement.

ON PEUT CHANGER LE
SEUIL D'ALERTE DANS
L'ONGLET **PARAMÈTRES**.





CONCLUSION :

TECHNIQUE/PERSONNEL